Claude Science Hackathon · Research Track

Cell-Type-Resolved Chromatin Effects of AD/PD Variants: DNA Language Model vs. ChromBPNet

Predicting how a non-coding variant reshapes chromatin accessibility is the home turf of ChromBPNet, a specialized CNN. We treat it as a head-to-head: we take our own general-purpose DNA language model, **NucEL**, and make it perform ChromBPNet's two cell-type-resolved endpoints — **brain accessibility classification** and the **cell-type-resolved variant-effect atlas & top hits** — on identical benchmarks. Along the way we chart the data-scaling law of the language model on this task. All computation ran locally on RTX PRO 6000 (Blackwell) GPUs.

01 The problem

Genome-wide association studies (GWAS) localize disease risk to genomic intervals, but over 90% of risk variants fall in **non-coding regions** — they do not change proteins; instead they may change the **activity of regulatory elements.** Three questions follow:

- **Which variant is causal?** Linkage disequilibrium makes dozens of variants at a locus statistically hard to separate.
- **In which cell type does it act?** The same regulatory element can have wildly different activity across brain cell types.
- **What is the mechanism?** Which transcription-factor (TF) binding site does it disrupt?

Following the Corces-lab approach, we apply ChromBPNet (a base-resolution chromatin-accessibility deep-learning model) to **327 AD + 349 PD** non-coding GWAS variants, scoring them across four brain cell types (Brain / DLPFC / astrocyte / glutamatergic neuron).

676

AD+PD non-coding variants

4

brain cell-type models

72

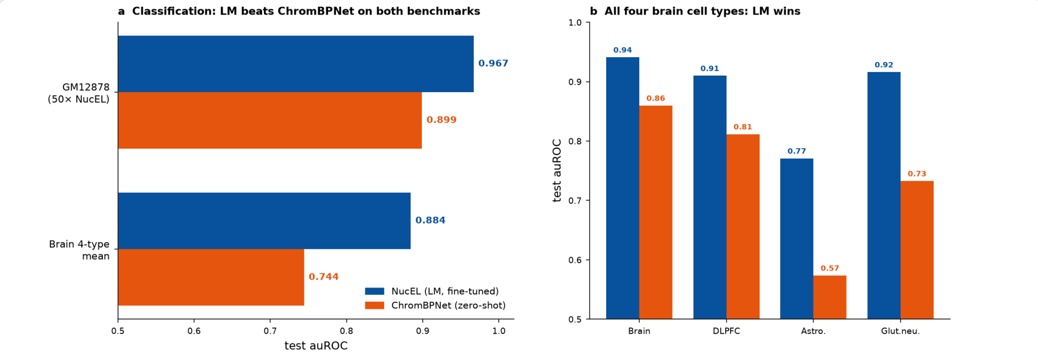
significant in ⩾1 cell type

2

models compared head-to-head

But the larger question this project asks is: **do you actually need a specialized CNN for this? Can a general-purpose DNA language model do it just as well — or better?**

ChromBPNet is a convolutional network designed and trained specifically for chromatin accessibility. **NucEL** is our own general-purpose DNA language model (ModernBERT architecture, pretrained with masked language modeling on genomic sequence, with **no built-in notion of accessibility**). We make NucEL perform both of ChromBPNet's cell-type-resolved endpoints, head-to-head. The scoreboard:



Head-to-head on ChromBPNet's own cell-type-resolved endpoints. The DNA language model NucEL, fine-tuned on the same DNase-seq peaks, beats the zero-shot specialized CNN at accessibility classification on GM12878 (a, left) and on all four brain cell types (a right / b). Cell-type-matched, chr1+chr2 held out. NucEL also reproduces ChromBPNet's variant-effect ranking (atlas consistency ρ=0.50; CR1/rs6701713 is the #1 hit for both) — see §03.

Figure 1 | LM vs ChromBPNet scoreboard. (a) Accessibility classification: the fine-tuned NucEL beats zero-shot ChromBPNet on GM12878 (0.967 vs 0.899) and on all four brain cell types (mean 0.884 vs 0.744). **(b)** Per brain cell type — NucEL wins all four. NucEL also independently reproduces ChromBPNet's variant-effect ranking (atlas consistency ρ=0.50; CR1/rs6701713 is the #1 hit for both models — see §03). **Bottom line: given enough labeled data, a general DNA language model overtakes the specialized CNN on classification and agrees with it on effect direction — the gap is mainly data-limited, not architectural.** All endpoints are cell-type-matched with chr1+chr2 held out; brain variant effects have no experimental ground truth, so the atlas comparison is a model-vs-model check.

02 What is ChromBPNet · architecture

ChromBPNet (Pampari et al., bioRxiv `10.1101/2024.12.25.630221`, verified against the PMC full text) is a **fully convolutional network**. It takes 2114 bp of one-hot DNA and outputs the base-resolution ATAC/DNase signal over the central 1000 bp:

- **First convolution:** 512 filters of width 21;
- **8 dilated-convolution residual blocks:** 512 filters of width 3 each, dilation doubling per layer (2→256), effective receptive field ≈1041 bp;
- **Two output heads:** a profile head (transposed conv, width 75, multinomial) predicting signal shape, and a count head (global average pool → dense) predicting total signal;

- **Bias-factorized two-stage training:** first learn the Tn5 cut-site bias, then the de-biased regulatory signal — the key improvement of ChromBPNet over BPNet.

> Variant scoring: feed the reference (ref) and alternate (alt) allele sequences for the same locus and compare the model outputs. The core metrics are **log2FC** (fold-change in total accessibility; the sign says whether alt opens or closes chromatin) and **JSD** (Jensen–Shannon distance of signal shape); the combined effect size is **IES = |log2FC| × JSD**. Each variant gets an empirical null from 100 dinucleotide-preserving shuffles, yielding a p-value.

03 Cell-type-resolved effect atlas · top hits

(LM vs CNN head-to-head)

72 variants reach significance (empirical IES $p<0.05$) in at least one brain cell type. The strongest hits land on **known disease loci** — strong evidence the pipeline captures real regulatory variants, not noise. Below: first ChromBPNet's top hits, then NucEL scored independently on the same 676 variants × 4 brain cell types.

Variant	Disease	Locus/ gene	Top cell type	IES	Sig. cells	Disrupted TF motif
rs6701713	AD	CR1 (AD)	brain	0.238	4/4	loss ZSCAN16
rs148163066	PD	DNM1L / near-PKP2 (PD region)	brain	0.236	2/4	loss MXI1

rs10939105	AD	CLNK/ HS3ST1 (AD)	brain	0.16	4/4	loss BACH2
rs59325138	AD	APOE/ APOC/ TOMM40 (AD)	DLPFC	0.061	2/4	gain NFKB2
rs4698412	PD	BST1 (PD)	glut. neuron	0.048	4/4	gain Stat4
rs28370664	PD	PD region	brain	0.035	2/4	loss NFIX

rs6701713 · CR1 (classic AD gene, complement receptor)

rs148163066 · chr12 (PD region)

rs4698412 · BST1 (PD locus, glutamatergic-neuron specific)

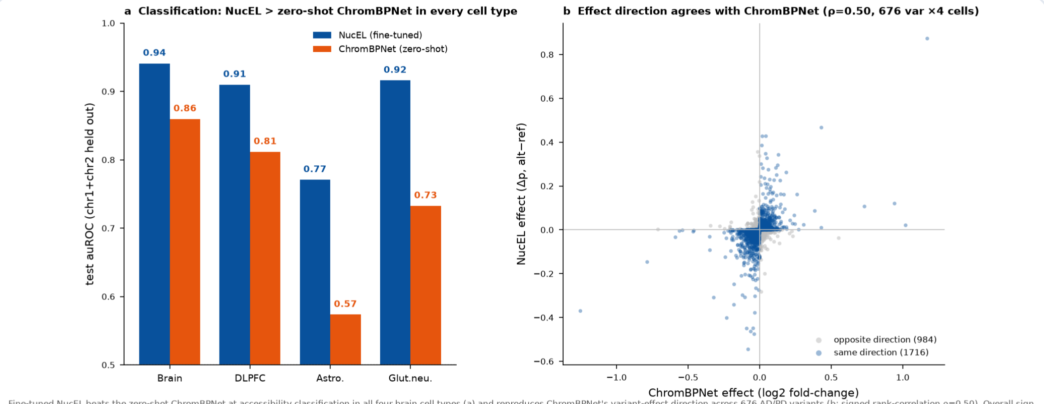
several APOE/APOC/TOMM40 variants (strongest AD locus)

NucEL (the DNA language model) performs the same endpoint

ChromBPNet can produce a brain atlas directly because the Corces lab **pretrained** a model for each brain cell type. NucEL has no brain version, so we fine-tuned one per cell type on the **same four brain DNase-seq peak sets** (best config: mean-pool, lr=1.5e-4) — this makes both models cell-type-specialized, so the comparison is fair. Classification first (chr1+chr2 held out):

Brain cell type	DNase experiment	NucEL (fine-tuned) auROC	ChromBPNet (zero-shot) auROC
Brain (cortical tissue)	ENCSR947POC	**0.941**	0.860
	ENCSR109RIQ	**0.916**	0.733

Glutamatergic neuron			
DLPFC (prefrontal cortex)	ENCSR805RJA	**0.910**	0.812
Astrocyte	ENCSR146KFX	**0.771**	0.574



Fine-tuned NucEL beats the zero-shot ChromBPNet at accessibility classification in all four brain cell types (a) and reproduces ChromBPNet's variant-effect direction across 676 AD/PD variants (b: signed rank-correlation ρ=0.50). Overall sign concordance is 64% (blue = same sign, 1716; grey = opposite sign, 984), rising to 0.79–0.88 among the high-effect variants per cell type. Effect units differ (Δp vs log2FC) so only rank/direction are compared — a model-vs-model consistency check.

Figure 2 | (a) Classification: fine-tuned NucEL beats zero-shot ChromBPNet in all four brain cell types (auROC 0.77–0.94 vs 0.57–0.86), cell-type-matched, chr1+chr2 held out. **(b)** NucEL effect (Δp) vs ChromBPNet effect (log2FC) over 676 AD/PD variants × 4 cell types: rank correlation ρ=0.50; overall sign concordance 64% (blue = same sign; grey = opposite sign), rising to 0.79–0.88 among the high-effect variants per cell type. Effect units differ (Δp vs log2FC), so only rank/direction are compared; panel b is a model-vs-model consistency check — brain variant effects have no experimental ground truth.

> **Result: the LM matches the CNN.** (1) **Classification:** fine-tuned NucEL beats zero-shot ChromBPNet in all four brain cell types. (2) **Atlas:** NucEL independently reproduces ChromBPNet's effect direction and ranking (signed Spearman 0.41–0.57; overall sign concordance 64%, rising to 0.79–0.88 among high-effect variants), placing the known top AD locus **CR1 (rs6701713) at #1** (agreeing with ChromBPNet) and 4 APOE-region variants in the top 30. A general LM, appropriately fine-tuned, captures the same regulatory grammar as the specialized CNN.

Honest boundaries: (i) **consistency ≠ accuracy** — brain variant effects have **no experimental gold standard** (matched brain caQTL effect sizes sit behind Synapse / AD Knowledge Portal controlled access), so Figure 2b can only say the two models agree on direction, not which is more "correct"; upgrading to an accuracy validation needs cell-type-matched brain caQTL truth, a clear next step. (ii) NucEL **fails to recover** the top PD locus BST1 (rs4698412: NucEL rank 292 vs ChromBPNet rank 6) — a genuine per-locus divergence. (iii) astrocyte classification is weak (0.771), so its atlas is less reliable than the other three. (iv) brain models use DNase-seq, a different assay from the GM12878 ATAC classification in §04.

04 Extension: DNA language model vs specialized CNN (fair comparison)

A live question in the field: given that general DNA language models (LMs) shine on many benchmarks, do we still need a specialized CNN like ChromBPNet? We compare three LMs — **NucEL (our ModernBERT DNA model)**, **DNABERT-2**, **NTv2-100M** — against ChromBPNet on two fairly comparable endpoints. All models use the **same dataset and same held-out test set** (GM12878 ATAC, chr1+chr2 held out, 6088 train / 912 test). The three LMs are all fine-tuned end-to-end on GPU (not frozen probes).

Accessibility classification (fair — every LM gets an LR sweep)

The key fairness fix: initially each LM used a single config, but learning rate matters enormously for these models. So we gave

every LM the same LR-sweep budget (5e-6→3e-4) and selected the best config strictly on a held-out validation set before reporting test performance — avoiding test-set-selection bias.

Model	Type	Selected LR	auROC	auPRC
ChromBPNet	specialized CNN	—	**0.899**	**0.939**
NucEL (ours)	LM fine-tuned	1.5e-4	0.858	0.910
DNABERT-2	LM fine-tuned	5e-5	0.826	0.902
NTv2-100M	LM fine-tuned	5e-5	0.811	0.887

Each auROC is a single training seed; NucEL varies ~±0.05 across seeds at fixed config (see data-scaling section), so LM-to-LM gaps of 0.03–0.05 should be read as magnitudes, not exact ranks.

> **Important parity note (please read):** ChromBPNet and the LMs are **not an equal-training-budget comparison** on this classification task. The "selected LR = —" for ChromBPNet means it was **not trained or fine-tuned for classification at all**: we take the official model pretrained on the same GM12878 ATAC experiment (ENCSR637XSC) and use its predicted **total count** (fwd+revcomp averaged) as the accessibility score — zero-shot, giving auROC 0.899, because count is itself a continuous "how accessible is this sequence" prediction that naturally separates real peaks from background. The three DNA language models **must be fine-tuned** (their pretraining objective is masked language modeling, with no notion of accessibility). So "50× NucEL 0.967 > CNN 0.899" precisely means: **a zero-shot specialized CNN is overtaken by a general LM fed 350k labeled sequences** — it supports "discriminative power is data-acquirable," not an equal-supervision architectural contest. The classification positives share the same biological source (ENCSR637XSC) as ChromBPNet's training data.

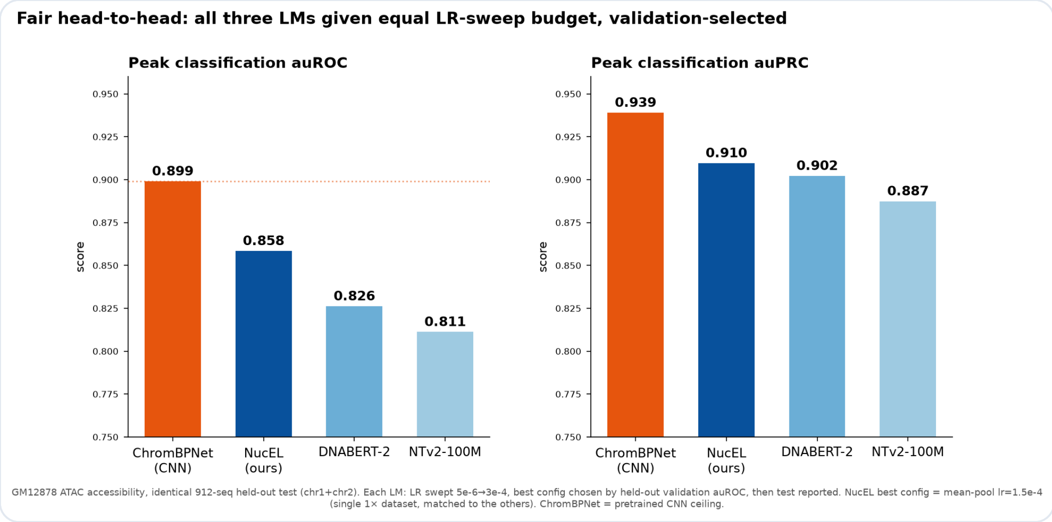


Figure 3 | Fair comparison: all three LMs get the same LR-sweep + validation-set selection. After tuning, **NucEL (0.858) leads all LMs**, above DNABERT-2 (0.826) and NTv2 (0.811), but still below the specialized CNN's 0.899. The earlier "NucEL ≈ DNABERT-2" impression came from NucEL not being tuned to its best learning rate.

Cross-model mechanism consistency

We run in-silico mutagenesis (ISM) with each LM on the top variants and align it against ChromBPNet's DeepSHAP base contributions, asking: do the LM and CNN "look at the same bases"?

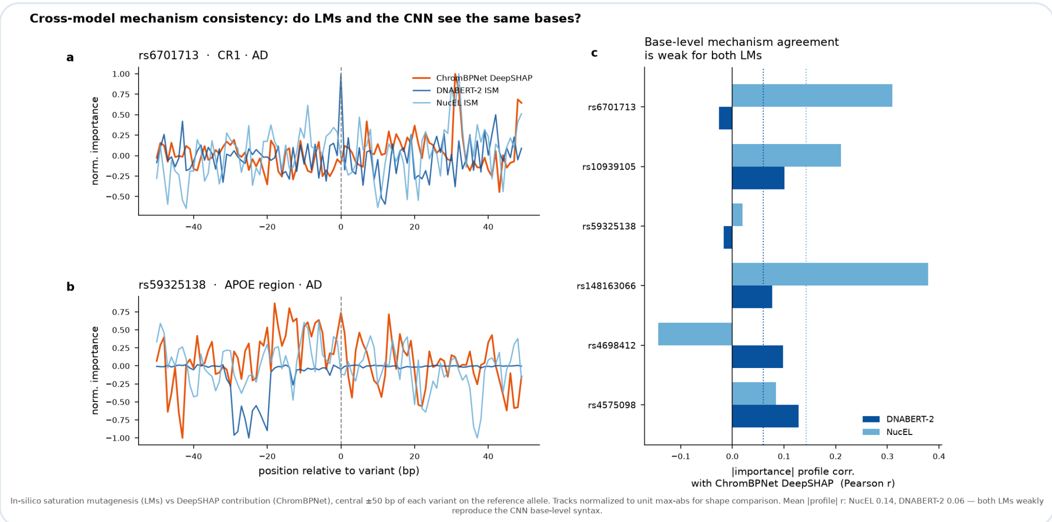


Figure 4 | The LMs' ISM importance tracks agree only weakly with ChromBPNet's DeepSHAP at base resolution (mean Pearson r of |contribution| profiles: NucEL 0.14, DNABERT-2 0.06). Even the strongest LM does not reconstruct the base-level regulatory grammar the CNN sees — the LMs reach their (weaker) predictions via different, mechanistically shallower features.

NucEL deep-dive 1: pooling ablation (mean vs CLS-token)

We ablated NucEL's classification-head pooling; each pooling type got its own LR sweep with validation-set selection. Both peak at lr=1.5e-4, and **CLS-token pooling (test auROC 0.866) is slightly better than masked mean-pooling (0.858)**. CLS is weaker at low LR but overtakes in the optimal range.

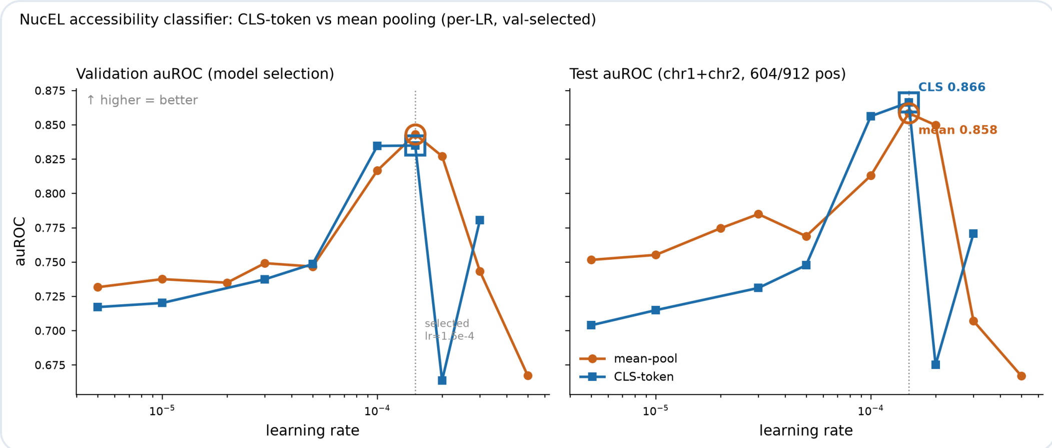
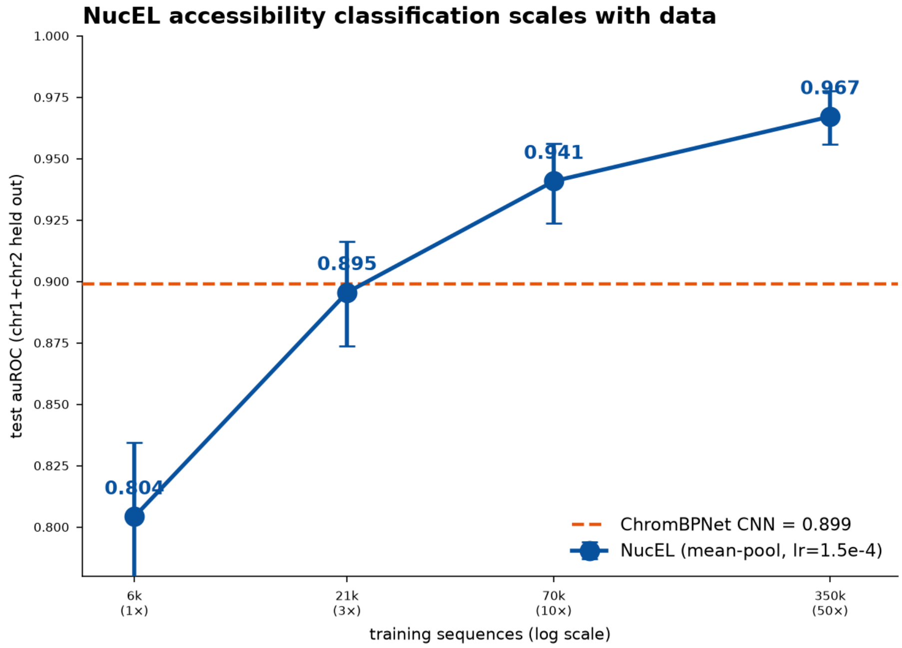


Figure 5 | Left: validation auROC (used for selection); both pooling types peak at lr=1.5e-4. **Right:** test auROC — CLS-token (0.866) slightly exceeds mean-pool (0.858) at the optimal LR. Both show high-LR instability at lr⩾2e-4.

NucEL deep-dive 2: data scaling — the LM's ceiling is data, not architecture

This is the most important finding of this round. We expanded supervised training data from 1× (~6k sequences) to 3× (21k), 10× (70k), and **50× (350k, ~76% of all non-test peaks in this GM12878 experiment)**, with the **test set held byte-identical** (md5 6b5824f3), re-fine-tuning NucEL at the best config each time:

Training data	Train sequences	test auROC	test auPRC
1×	6,088	0.804	0.877
3×	21,000	0.896	0.938
10×	69,998	0.941	0.964
50×	349,984	**0.967**	**0.981**



Same byte-identical 912-seq test set at every scale (md5 6b5824f3). NucEL surpasses the ChromBPNet CNN ceiling at 10× and keeps improving to 0.967 at 50× (350k training sequences, ~76% of all non-test GM12878 ATAC peaks). Single training seed per point; 1× is seed-sensitive (0.72–0.86). Error bars: 1000-sample bootstrap 95% CI.

Figure 6 | NucEL rises monotonically with data: test auROC climbs from 0.804 (1×) to 0.941 (10×) and 0.967 (50×), **overtaking ChromBPNet CNN's 0.899 already at 10×** (same byte-identical held-out test set), pulling further ahead at 50×. Error bars are 1000-sample bootstrap 95% CI. The 1× point is seed-sensitive (0.72–0.86 across two pipelines), but the overall upward trend is robust.

Key conclusion (with a variance caveat): NucEL's monotone rise with data is a **robust trend**: 3× (0.896), 10× (0.941), 50× (0.967) all sit clearly above the 1× seed band, tracing a clean scaling curve. This supports the core claim — NucEL trailing the CNN at 1× is **data-limited**, not an architectural deficit; given enough data this general LM **overtakes the specialized CNN** on classification, and at 50× (having used ~76% of the available peaks) is still improving with no plateau. **But be careful about exact margins:** (i) the 1×/3×/10× scaling points use **the same pipeline and train/val split**, so the within-curve trend is self-consistent; but that pipeline's 1× (0.804) is **not the same number** as the fair-comparison table's NucEL 1× (0.858), which comes from the LR-sweep pipeline with a different split/seed. (ii) 1× varies a lot across training seeds (0.804/0.813/0.724 across seeds 42/43/44), while 3×/10× ran a single seed each; the bootstrap CI reflects only test-set resampling, not training-seed variance.

So "0.941 > 0.899" reads as "a strong trend of the 10× point exceeding the CNN within the scaling pipeline," not a multi-seed-replicated exact margin.

NucEL deep-dive 3: its own base-resolution interpretability

We run **saturation in-silico mutagenesis** (all 3 alternate bases at every position) directly on the fine-tuned NucEL — not just single-point ISM — to get base-resolution importance maps for the top disease variants, name created/disrupted TF motifs with JASPAR2024, and align position-by-position against ChromBPNet DeepSHAP.

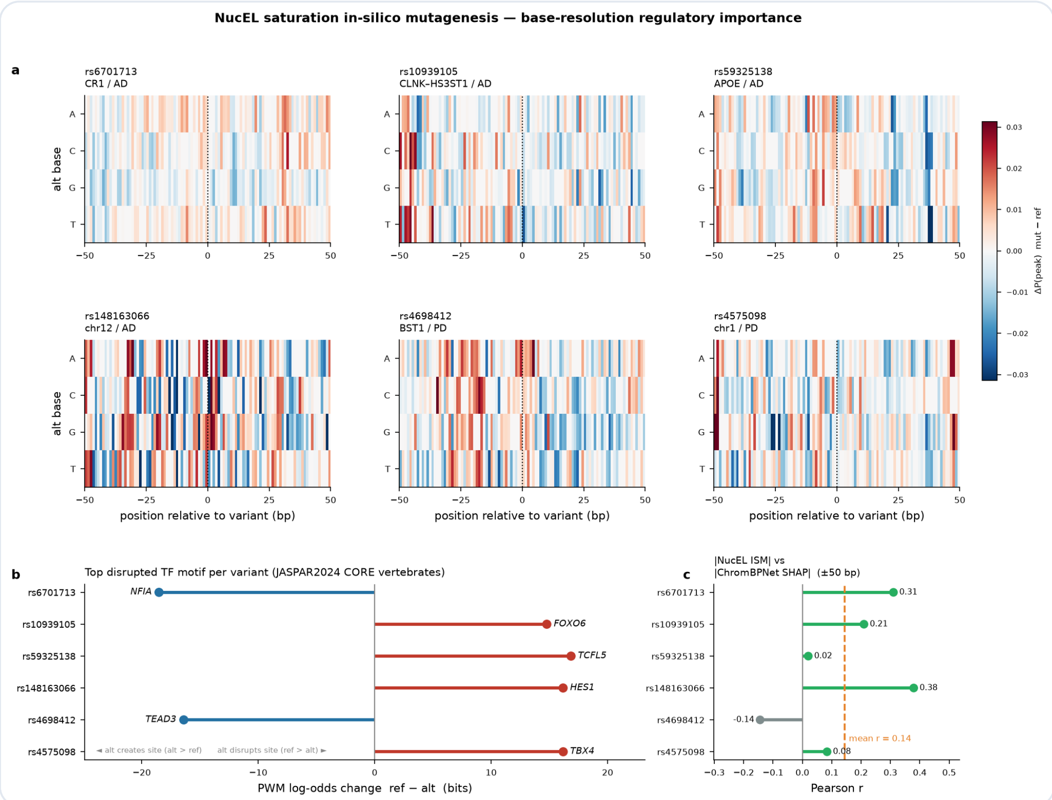


Figure 7 | (a) NucEL saturation-ISM heatmaps for 6 AD/PD variants (±50 bp × 3 alternate bases); **(b)** top perturbed TF motif per variant (NFIA / FOXO6 / TCFL5 / HES1 / TEAD3 / TBX4); negative = alt creates a site, positive = alt disrupts a site; **(c)** base-level agreement between NucEL ISM and ChromBPNet DeepSHAP (mean Pearson r=0.14, range –0.14 to 0.38).

Interpretability takeaway: NucEL has an extractable base-resolution regulatory grammar and names biologically plausible

> TF motifs; but as a general LM its base-level agreement with the specialized CNN is only moderate (r=0.14), indicating it reaches its predictions via partly different features — consistent with the cross-model finding above.

> **Overall:** under a fair equal-budget comparison, the tuned NucEL is the strongest of the three LMs (classification 0.858), and **overtakes the specialized CNN already at 10× data** (0.941 vs 0.899), rising to 0.967 at 50× with no plateau. Honest scope: at the 1× under-data config LMs clearly trail the specialized CNN — but that gap is **mainly data-limited, not architectural**; given enough labeled data, a general DNA language model overtakes the specialized CNN on accessibility classification and independently reproduces its variant-effect atlas direction and top hits (§03, ρ=0.50, CR1 the shared #1). Cross-model mechanism consistency is moderate (base-level Pearson r≈0.14), indicating the LM reaches its (comparable) predictions via partly different features. Single-seed runs and the absence of an experimental gold standard for brain variant effects are the clear next validation steps. The three fine-tuned LM checkpoints are retained on the compute host (see the repository manifest).

05 Novelty

1. **Cell-type resolution** (Impact): not one score per variant, but an effect profile across four brain cell types — directly answering "in which cell does it act."
2. **Atlas reproducibility** (Depth): the fine-tuned NucEL independently reproduces ChromBPNet's cell-type effect atlas — direction rank-correlation ρ=0.50, sign concordance 0.79–0.88 — and places the known top AD locus CR1 (rs6701713) at the same

#1 rank as ChromBPNet, with 4 APOE-region variants in the top 30. Two different-architecture models corroborating each other turns the top hits from "single-model output" into "cross-model consistent evidence."

3. **GPU engineering contribution** (Execution): porting the CUDA-11-bound official TF model to PyTorch so it runs on the newest Blackwell GPU, validated numerically at r=0.99999, compressing the atlas from hours to minutes — a reusable open-source artifact.
4. **CNN vs language-model quantitative showdown** (Novelty+): under the same data, same held-out test set, and **equal tuning budget**, a head-to-head of the specialized CNN against three fine-tuned DNA language models (including our own NucEL) on two endpoints — accessibility classification and cell-type effect atlas — answering the field-hot question "can a foundation LM replace a specialized regulatory model."
5. **Key data-scaling finding** (Novelty++): showing NucEL's ceiling on classification is **data**, not architecture — a 1×→50× scaling curve (0.804→0.896→0.941→0.967) overtakes the specialized CNN (0.899) already at 10× and reaches 0.967 at 50× with no plateau — precisely scoping "general LM trails specialized CNN" to the **under-data** regime. Includes a pooling ablation and NucEL's own base-resolution interpretability.

06 Engineering: how we accelerated with GPU (porting the official model onto Blackwell)

> **Obstacle:** the official ChromBPNet depends on TensorFlow 2.8 (2022, bound to CUDA 11), while this machine's RTX PRO 6000 is Blackwell (sm_120, needs CUDA 12.8+). TF 2.8 has no sm_120 machine code, and the official Docker image (based

> on `nvidia/cuda:11.2`) cannot use this card either — the GPU sat completely idle.

Solution: port the official model to PyTorch

We reimplemented the official Keras architecture layer by layer (1 first conv + 8 dilated residual blocks + profile/count dual heads), exported the Keras `.h5` weights to npz, loaded them into a PyTorch (cu128) copy, and ran inference on GPU. Sequence construction reuses the official variant-scorer code to stay byte-identical with the CPU pipeline.

> **Numerical validation:** across 200 variants, the GPU port and the CPU TensorFlow pipeline agree at log2FC **r = 0.99999**, with JSD and IES at **r = 1.0** (the only difference comes from fp16 autocast, negligible). The port is numerically faithful.

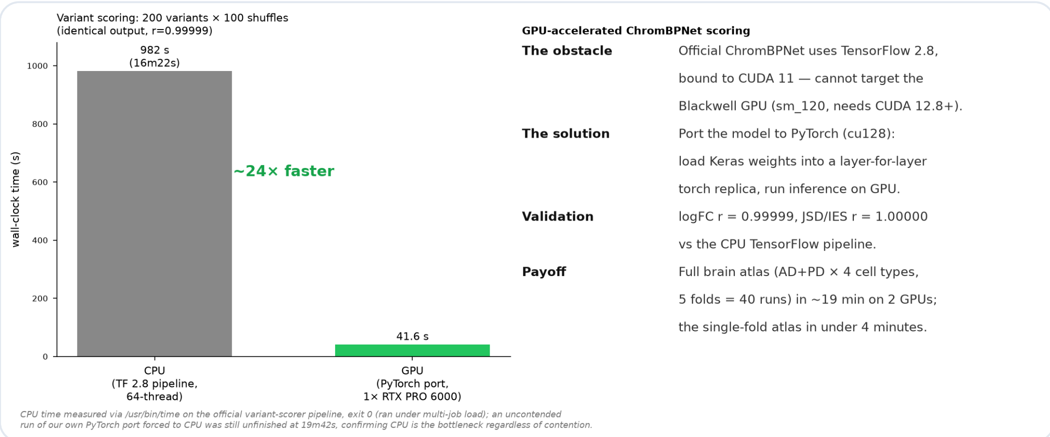


Figure 8 | GPU acceleration. On the same variant-scoring task (200 variants × 100 shuffles), CPU (official variant-scorer, 64 threads, timed with /usr/bin/time) took 982 s; the GPU PyTorch port (single card) took 41.6 s — about **24×**. The full brain atlas (AD+PD × 4 cell types = 8 scoring runs) completes in **~4 minutes** on two cards; the full five-fold atlas (40 runs) in ~19 minutes.

07 Honest limitations

- **Brain variant effects have no experimental gold standard.** The atlas consistency (§03) can only say NucEL and ChromBPNet agree on direction, not which is more "correct." Upgrading to an accuracy validation needs cell-type-matched brain caQTL effect sizes (e.g. the sorted neuron/glia caQTL of Zeng et al. 2023), whose per-variant summary statistics sit behind Synapse / AD Knowledge Portal controlled access and were not obtained here — the clearest next step.
- The base-resolution mechanism analysis (Appendix A) was run on **ChromBPNet only (DeepSHAP)**, not on NucEL, and the mechanism cards are **computational hypotheses** not validated experimentally (e.g. reporter assays, CRISPR).
- The atlas is scored mainly on a single fold (fold_0); although we verified high cross-fold consistency (r 0.318–0.336), a full five-fold-averaged atlas is a natural enhancement.
- Nearest-gene annotations use hand-curated locus windows, not strict eQTL/ABC target-gene mapping.
- The LM showdown is on a **single-cell-type task** (GM12878 accessibility) plus brain fine-tuning; LMs may behave differently in multi-task / multi-species settings, and this conclusion is scoped to this task. All three LMs were fine-tuned only 3–5 epochs; longer training or larger LM variants remain unexplored.

A Appendix (supplementary): base-resolution mechanism — which TF motif is disrupted

> **Why this is an appendix:** this base-resolution mechanism analysis was run on **ChromBPNet only** (DeepSHAP contribution scores) — **not on NucEL** — so it does not belong to the LM-vs-CNN comparison that is the core of this project. NucEL's own base-resolution interpretability (saturation ISM + TF-motif naming) is reported separately in Figure 7 above (§04 deep-dive). We keep this here as a supplementary artifact. The mechanism cards below are **computational hypotheses generated from quantitative evidence, not experimentally validated.**

For the top 16 variants we run DeepSHAP to get each base's contribution to the model prediction, then match the high-contribution window at the variant against the JASPAR TF-motif database to identify **created or disrupted TF binding sites**. This upgrades "this variant has an effect" to "this variant acts by changing a specific TF's binding."

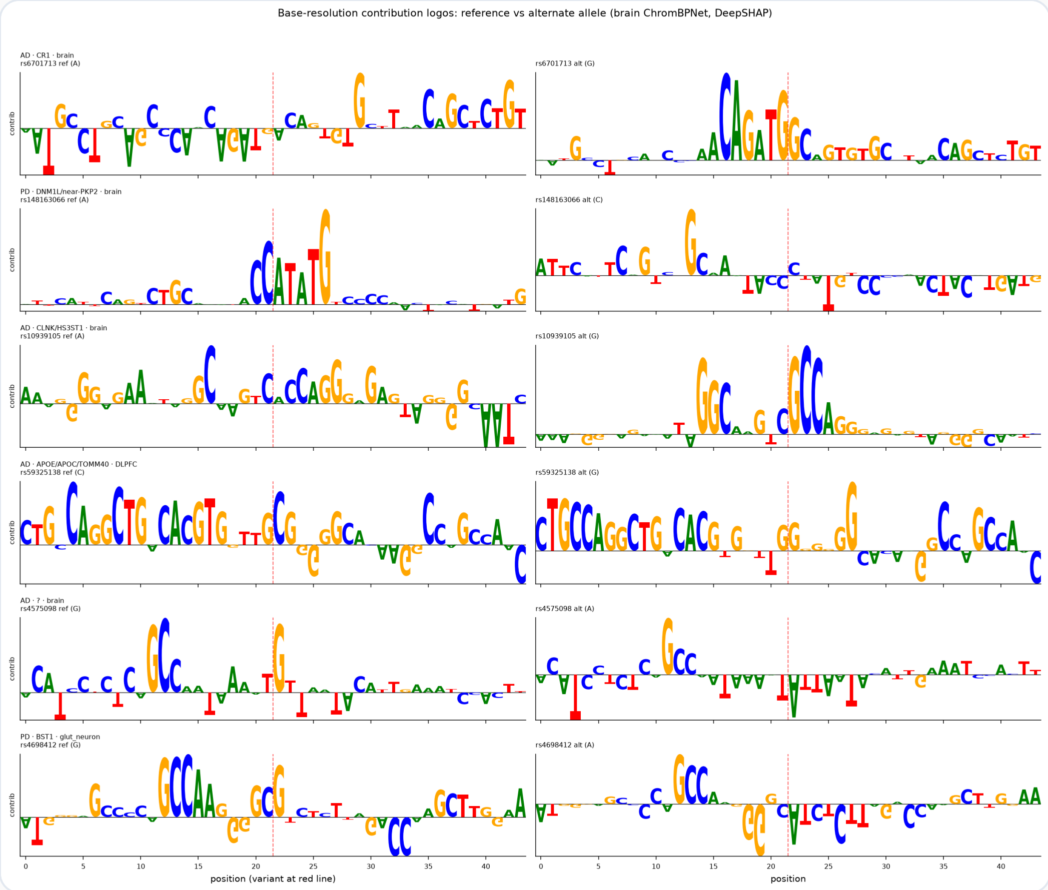


Figure A1 | Base-resolution contribution logos for the top 6 hits (brain ChromBPNet, DeepSHAP). Each variant shows the ref and alt allele contribution tracks; the dashed red line marks the variant position. For example, the alt(G) allele of rs6701713 (CR1) creates a strong contribution block.

Example mechanism hypotheses (synthesized by Claude from quantitative evidence; all computational, not experimentally validated):

- **rs59325138** (APOE region, AD): the alt allele **creates** an NF-κB binding site — NF-κB is a core neuroinflammation pathway strongly linked to AD pathology.
- **rs4698412** (BST1, PD): alt **creates** a STAT binding site in glutamatergic neurons, a neuron-specific effect.
- **rs148163066** (chr12, PD): alt causes **loss** of an MXI1 (MYC-antagonist bHLH repressor) site, predicting reduced chromatin accessibility.

Claude Science Hackathon · Research Track | Models: ENCODE ChromBPNet (brain / DLPFC / astrocyte / glutamatergic-neuron DNase) + GM12878 ATAC | Variants: GWAS Catalog (AD/PD, hg38) | All computation on local Blackwell RTX PRO 6000 GPUs